

9) Pin Description of MN6617

● IC301: MN6617 (Digital Signal Processing: EFM Decoder, Error Correction, CLV Servo)

Pin No.	Mark	I/O Division	Function	Pin No.	Mark	I/O Division	Function
1	BLKCK	O	Sub-code block (Q data) clock (75Hz)	33	RESY	O	Resynchronizing signal (Not used, open)
2	CLDCK	O	Sub-code frame (Q data) clock (7.35kHz)	34	FCLV	O	Synchronizing detection signal (Not used, open)
3	SUBQ	O	Sub-code (Q data) output	35	IPBYTE	O	Interpolation flag for each byte (Not used, open)
4	CRC	O	Sub-code (Q data) CRC check (Not used, open)	36	IPSEL	I/O	Interpolation prohibited (Not used, connected to GND)
5	RST	I	Reset signal input (reset at "L")	37	FLAG5	O	C2 decoder correction flag 3 (Not used, open)
6	MLD	I	Command load input	38	FLAG4	O	C2 decoder correction flag 2 (Not used, open)
7	MCLK	I	Command clock input	39	FLAG3	O	C2 decoder correction flag 1 (Not used, open)
8	MDATA	I	Command data input	40	FLAG2	O	C1 decoder correction flag 2 (Not used, open)
9	DMUTE	I	Muting control (Not used, connected to GND)	41	FLAG1	O	C1 decoder correction flag 1 (Not used, open)
10	TRON	I	Tracking servo ON signal (tracking servo ON at "L")	42	FLCK	O	Crystal frame clock
11	STAT	O	Processing condition (CRC, OTC, CLVOK, TT STOP) output	43 50	D7 D0	I/O	16K RAM data output
12	SMCK	O	Clock output (4.2336MHz)	51	RAMOE	O	16K RAM $\overline{OE}$ signal
13	PMCK	O	Pitch control clock output (Not used, open)	52	RAMWE	O	16K RAM $\overline{WE}$ signal
14	ITC	I	Track counter input signal (Not used, open)	53 63	RAMA0 RAMA10	O	16K RAM address signal (RAMA0: LSB, RAMA10: MSB)
15	TEST	I	Test mode selection (Not used, connected to +5V)	64	PC	O	Spindle motor ON signal (ON at "L")
16	X2	O	Clock output (Not used, open)	65	EC	O	Spindle motor drive signal
17	X1	I	Clock input	66	FG	I	Spindle motor FG signal input (Not used, open)
18	SEL	I	DA output parallel/serial selection (serial at "L") (Not used, connected to GND)	67	VCNT	—	(Not used, open)
19	LDG/WDC	O	L channel deglitch signal/serial data word clock. (Not used, open)	68	REXT	—	(Not used, open)
20	RDG	O	Clock signal output for spindle motor control (Not used, open)	69	VDD	I	Power supply (+5V)
21	DEMPH	O	De-emphasis ON signal (de-emphasis ON at "H") (Not used, open)	70	PD	—	(Not used, open)
22	IPFLAG	O	Interpolation flag (interpolation at "H") (Not used, open)	71	PCKO	—	(Not used, open)
23	FLAGO	O	Error flag (error at "H") (Not used, open)	72	PCK	I	PLL extract clock input
24	FLAG6	O	16K RAM address reset signal (reset at "H") (Not used, open)	73	VDD	I	Power supply (connected to +5V)
25	XCK	O	Clock (16.9344MHz) output (Not used, open)	74	EFM	I	EFM signal input (PLL)
26	SRDATA	O	Serial data output (MSB first)	75	SRF	I	EFM signal input (DSL)
27	SRDATA	O	Serial data output (LSB first) (Not used, open)	76	DO	I	Drop-out signal (Drop-out at "H")
28	SRCK	O	Serial data beat clock.	77	CLVS	O	11T servo OK signal (OK at "H") (Not used, open)
29	WDCK	O	Serial data word clock (Not used)	78	FPC	O	PLL frequency comparison signal
30	BYTCK	O	Serial data byte clock (Not used, open)	79	BSEL	O	PLL frequency in take operation signal. (Not used, open)
31	GND	I	GND terminal	80	SRFO	—	(Not used, open)
32	R/L	O	R/L signal	81	NSRF	—	(Not used, open)
				82	RF	—	(Not used, open)
				83	SUBC	O	Sub-code serial output data (Not used, open)
				84	SBCK	I	Clock for sub-code serial output (Not used, open)

Table 18-5

10) Block Diagram of Digital Circuit

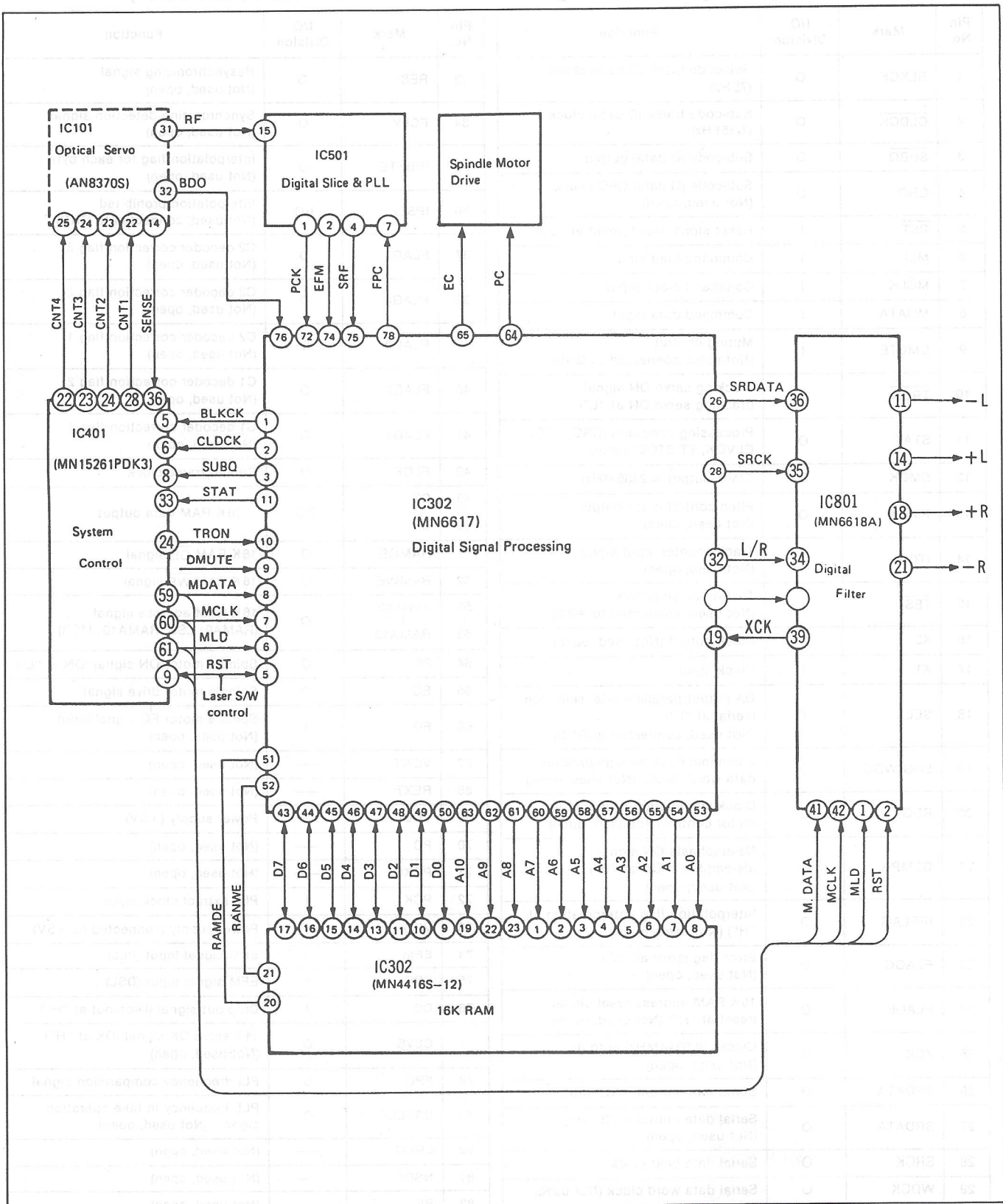


Figure 18-16

3) The Effect of the Sampling Frequency

Another important factor is sampling frequency, for efficiency in digital operations, this is kept low (44.1KHz) at a little over twice the maximum audio frequency (20.0KHz). During the sampling process this frequency and its harmonics or multiples become modulated with the original waveform in a series of frequency bands as shows figure 19-2.

The bottom end of the lowest band is rather close to the top of the audio frequency range.

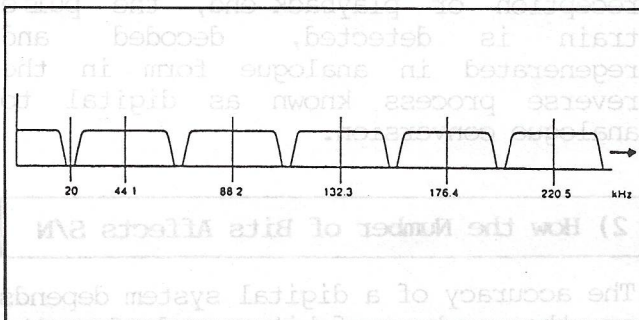


Figure 19-2

Figure 19-2 shows the frequency spectrum produced by sampling showing the periodic repetitions of the audio band. The bottom end of the lowest band is close to the audio frequency range.

4) Quantization Noise

In a digital audio system the audible noise has a different character to the familiar noise in an analogue audio system. The digital audio system background noise is called quantization noise.

The quantization noise of a system can be defined as the error signal between the original waveform before the analogue to digital conversion process, and the restored waveform after the digital to analogue conversion process. This error signal can be considered as both noise and distortion at the same time.

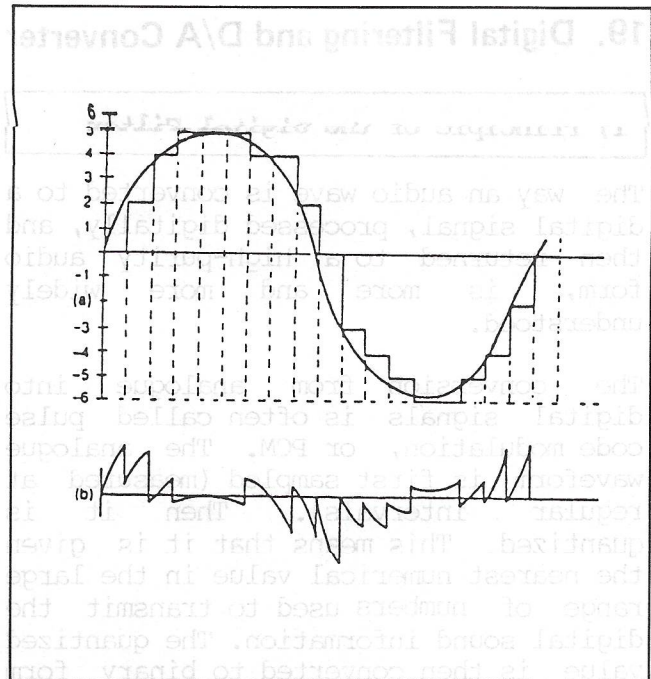


Figure 19-3

Figure 19-3 shows the quantization noise, The difference between the original audio waveform and the regenerated wave after D/A conversion (a) causes quantization noise (b).

The 16 bits analogue to digital conversion at the recording stage delivers a 16 bits digital signal with an intrinsic accuracy or signal to noise ratio of maximum 96dB.

An ideal digital to analogue converter (D/A) would deliver an analogue output signal without adding noise to it. However, ideal D/A do not exist a small but finite electronic inaccuracy adds noise to the quantization noise of the digital signal.

The inaccuracy of a 16 bits D/A (the resolution limit of half the least significant bit) can be interpreted as a noise source that reduces the signal to noise ratio of the D/A to 96dB.



Figure 19-1

10. Digital Filtering and D/A Converter

1) Principle of the Digital Filter

The way an audio wave is converted to a digital signal, processed digitally, and then returned to a high-purity audio form, is more and more widely understood.

The conversion from analogue into digital signals is often called pulse code modulation, or PCM. The analogue waveform is first sampled (measured at regular intervals). Then it is quantized. This means that it is given the nearest numerical value in the large range of numbers used to transmit the digital sound information. The quantized value is then converted to binary form and encoded into a pulse train representing the series of numbers. This process of analogue to digital conversion is shown in figure 19-1.

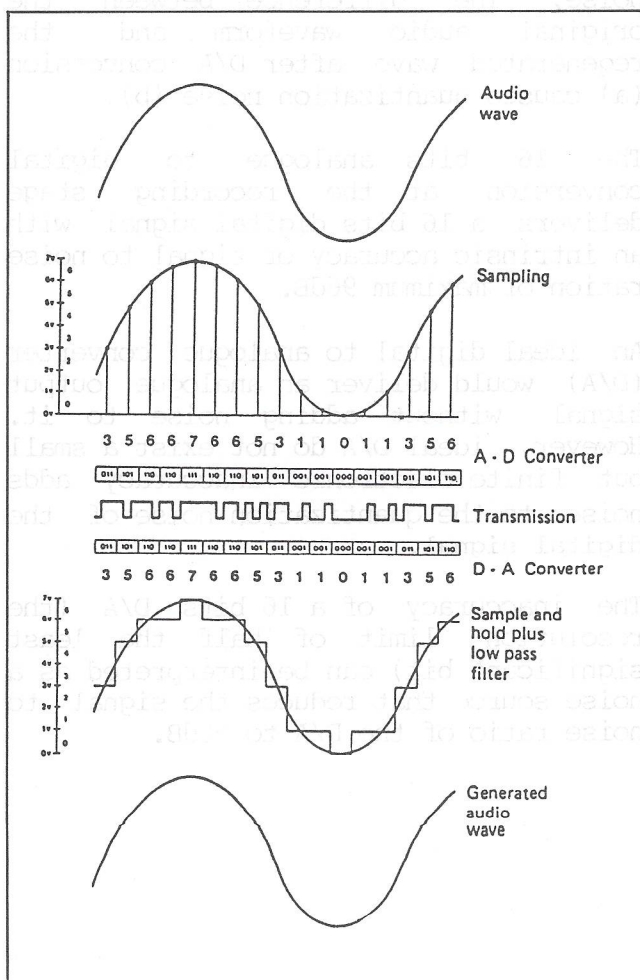


Figure 19-1

Figure 19-1 shows the digital sound reproduction process. The audio signal is sampled, quantized, converted to binary form, and encoded for recording or transmission.

Reversing the process produces a replica of the original signal. It is in this second stage that CD players use digital filtering to achieve exceptionally accurate reproduction.

The pulse train is fed into the transmission system (a cable or radio link, a tape, or a disc etc). At the reception or playback end, the pulse train is detected, decoded and regenerated in analogue form in the reverse process known as digital to analogue conversion.

2) How the Number of Bits Affects S/N

The accuracy of a digital system depends on the number of bits used for the binary number. This is because the number of bits determines the accuracy with which the samples taken from the original waveform can be quantized, i.e. expressed in digital form.

Two bits give four possible values for use in quantization, also three bits give eight possible values.

Every bit added doubles the number of possible values. In audio terms, this adds 6dB to the signal to noise ratio, because it reduces the quantization noise by a factor of 2.

The sixteen bits as used in the CD system give 65536 (2^{16}) possible values for quantization, and a signal to noise ratio of 96dB. The audio signals are encoded in the actual CD with this ratio.

96dB is approximation. An additional factor, which depends on the type of signal, is always present, regardless of the number of bits; for sine wave signals this makes the theoretical figure 97.8dB.

Figure 19-5 shows the generation of sample (a) and hold (b) produces a step waveform closer to the original.

6) How D/A Conversion Operate

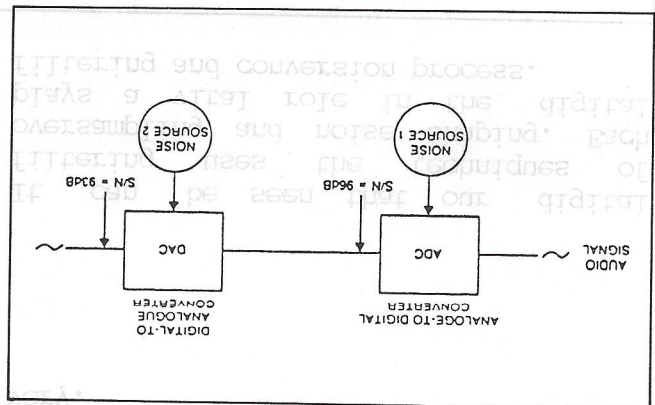
These frequency bands can overload the power amplifier, and mix with audio frequencies or other signals to cause audible interference. They must be filtered out.

With 16 bits (65536 possible numbers) and a sampling frequency of 44.1kHz, the CD system is theoretically capable of far higher reproduction quality than any conventional record player. But to achieve this quality, the output signal that the player derives from the disc must contain only the original audio frequencies, and not the periodic repetitions of the audio frequencies produced by the sampling (see figure 19-2).

5) Need for Filtering

Figure 19-4 shows the signal to noise ratio deterioration in a digital chain, consisting of an A/D and a practical D/A.

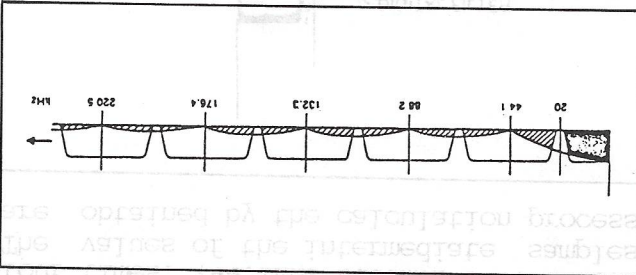
Figure 19-4



The effect of 16 bits analogue to digital conversion followed by 16 bits digital to analogue conversion with 1/2 LSB accuracy is to double the noise. In consequence, the maximum signal to noise ratio achievable in practice with a 16 bit system such as CD digital audio is 93dB.

Figure 19-6 shows the effect of sample and hold. The remaining signal components are shown shaded.

Figure 19-6



This sample and hold process is illustrated in figure 19-5 (b). It substantially reduces the frequency band components, as can be seen from figure 19-6. Significant components still remain, however, particularly in the region near the audio band.

The outline of this series resembles the audio wave, but the constant on and off switching produces the infinite series of frequency bands which was shown in figure 19-2. In practice, each sample is held in the converter until the next sample arrives, to produce a step waveform which is much nearer the shape of the audio waveform than the series of sample.

In the pulse train a series of binary number represents quantized samples of the original audio frequency. If these are simply converted to analogue values, a series of sample is produced (see figure 19-5).

Figure 19-5

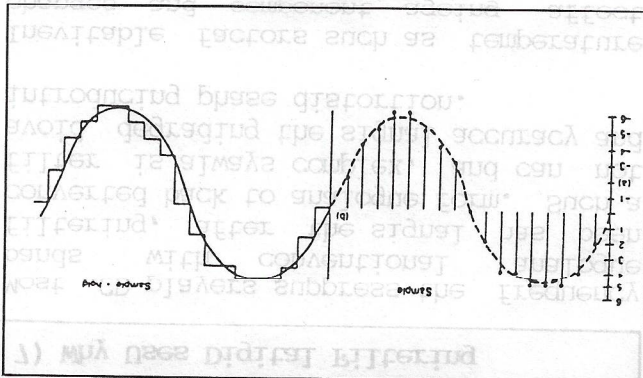
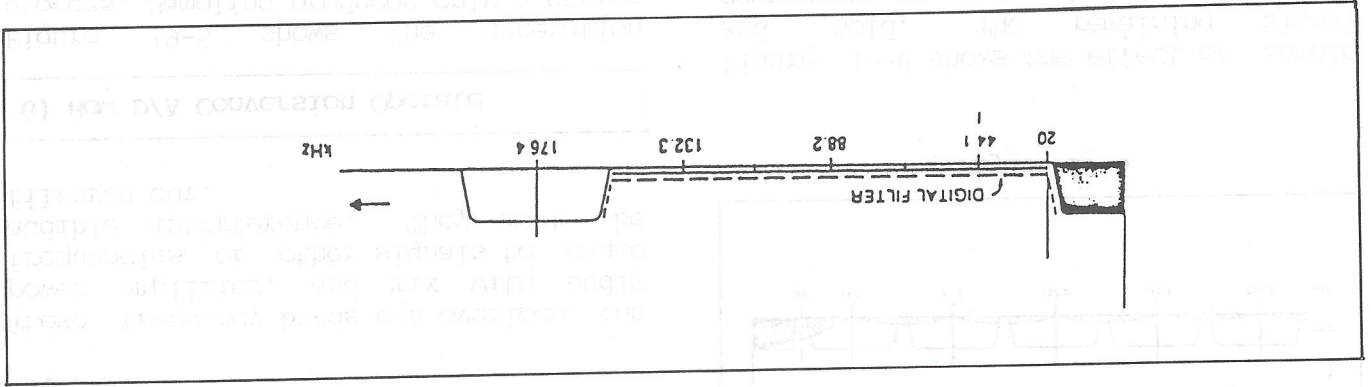


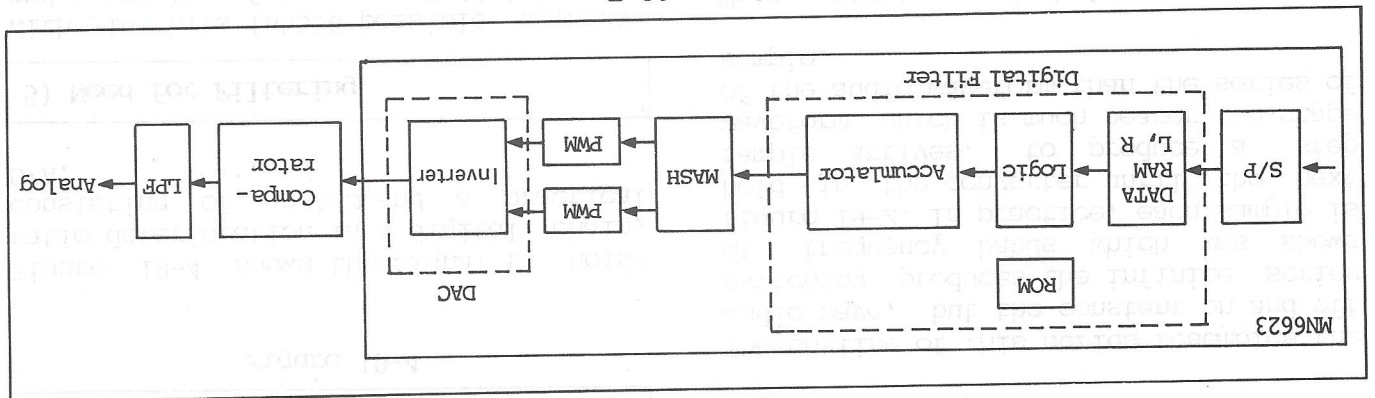
Figure 19-8



oversampling means multiplying the sampling frequency, in this case by four times. (44.1KHz becomes 176.4KHz.) The values of the intermediate samples are obtained by the calculation process

which determines the filter characteristic. An immediate effect of the oversampling as shown in figure 19-8 is to suppress all the lower frequency bands, centered on fs, 2fs, and 3fs.

Figure 19-7



A schematic diagram of the complete our D/A conversion system including digital filtering, is shows in figure 19-7.

8) Digital Filtering Process

Inevitable factors such as temperature changes and component ageing affect performance. Most CD players on the

Most CD players suppress the frequency bands with conventional analogue filtering, after the signal has been converted back to analogue form. Such a filter is always complex, and can not avoid degrading the signal accuracy and introducing phase distortion.

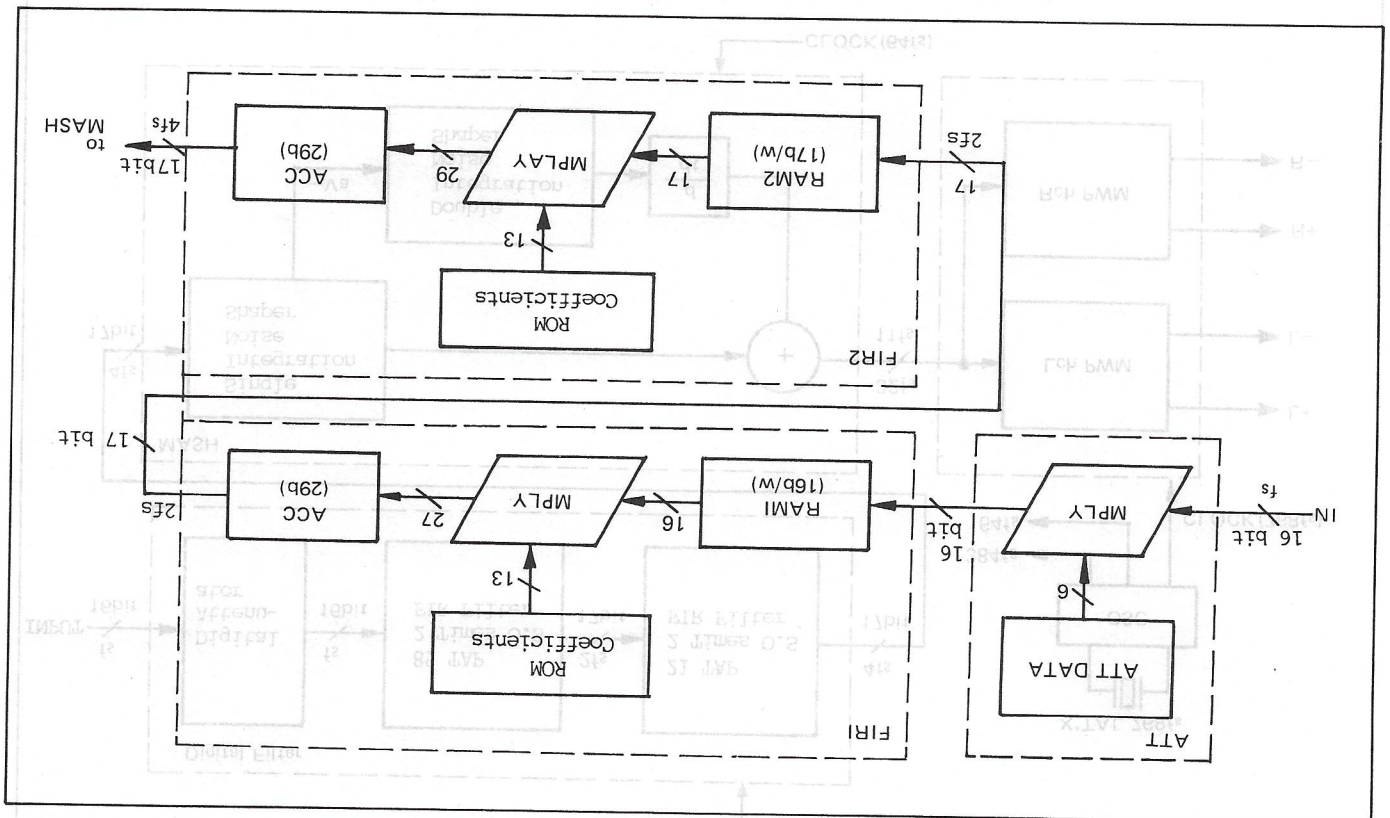
7) Why uses Digital Filtering

other hand, carry out this essential filtering process digitally, with no loss of accuracy, and phase distortion.

It can be seen that our digital filtering uses the techniques of oversampling and noise shaping. Each plays a vital role in the digital filtering and conversion process.

The digital filter block contains many memories (delay logics), as shown in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 19-11.

Figure 19-10 Digital Filter Block Diagram



The digital filter block consists of three sub-blocks: ATT, FIR1, and FIR2 as shown in figure 19-10. It performs digital attenuation and oversampling to convert the input 16-bits, fs data into a 16 bits, 4fs signal.

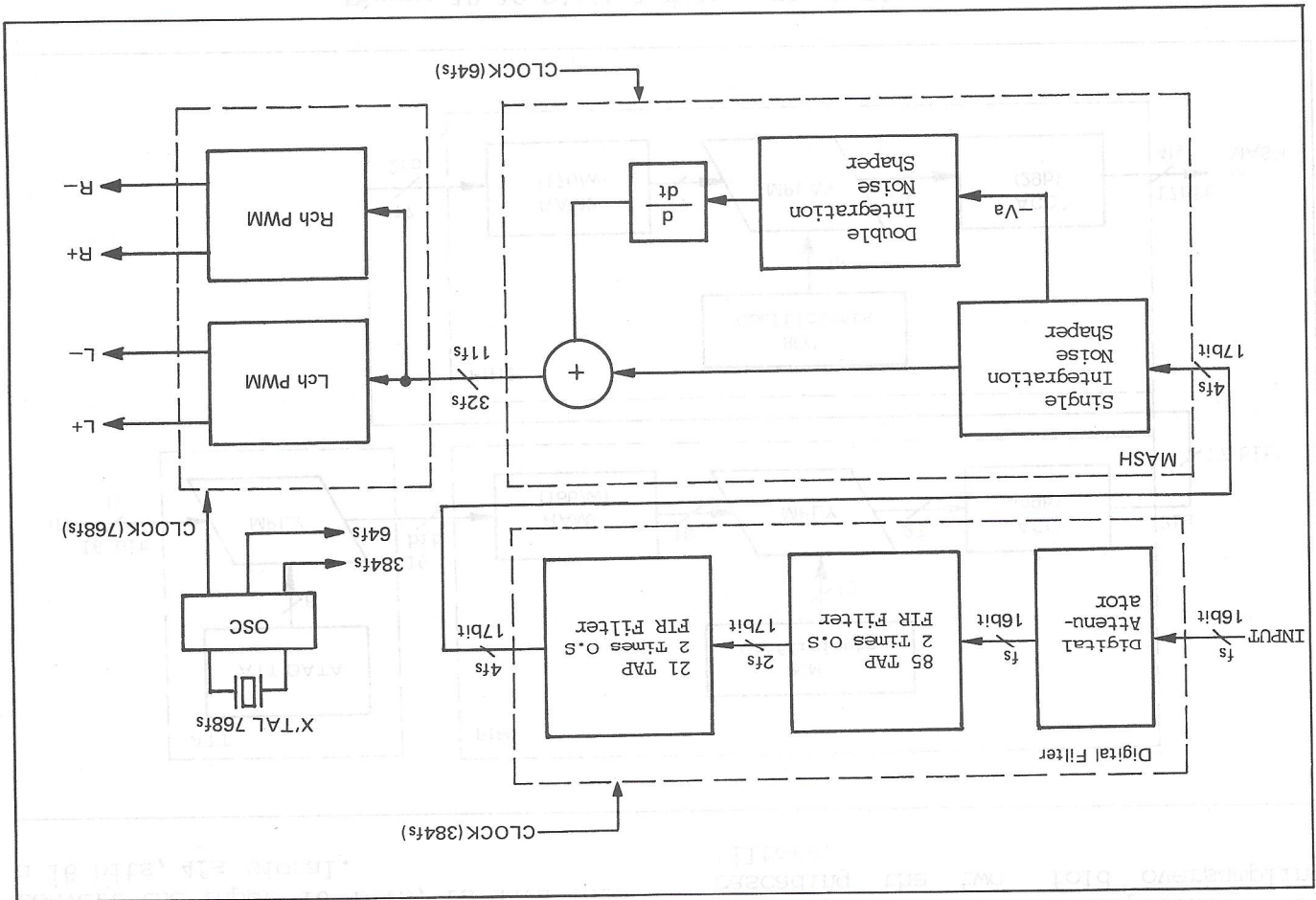
Digital Filter Block

The ATT attenuates the input data in a $1/64$ increment by multiplying the data by any 6 bits data. FIR1 and FIR2 are 2-fold oversampling digital filters with 85 and 21 taps each. Four-fold oversampling is accomplished by cascading the two fold oversampling filters.

The MASH block operates at 32fs, where fs is the sampled input signal frequency. This means that a 32 fold oversampling filter is required. On the MN6623, however, the first digital filter oversamples the fs signal into a 4fs signal. The following MASH block then performs zero-order holding on the 4fs signal to convert it into a 32fs signal.

The digital filter and MASH blocks use 384fs and 64fs clocks, respectively, to perform time-division multiplexing on the left and right channel signals. The PWM block uses a 768fs clock to process the left and right signals independently. The MASH block converts the 16 bits, 32fs signal into 11 values, which after Pulse-Width Modulation in the PWM block become analog signals. The following section describe the function of each MN6623 block.

Figure 19-9 MN6623 Block Diagram



The MN6623 D/A converter system is composed of a digital filter, MASH, and PWM blocks as shown in figure 19-9.

9) MN6623 Actual Operation